

ICS 2413: Artificial Intelligence

CampusMind — Part 1: AI Problem Definition (Week 1)

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A. Problem Statement

CampusMind is a campus navigation problem in which a student needs to move between six predefined campus locations. Given a starting location and a destination, the system should represent the campus as a formal search problem so that search algorithms can later find a valid route. Each connection has a movement cost represented by its distance in kilometres.

B. Initial State

The initial state is the location where the student begins the journey. For the required test scenario used later in the project, the student starts at the Main Gate (L1).

Initial state: CurrentLocation = Main Gate (L1)

C. Goal State

The goal state is the destination location the student wants to reach. The required test scenario uses the Science Laboratory (L4) as the destination.

Goal state: GoalLocation = Science Laboratory (L4)

D. State Representation

A state is represented by the student's current campus location. The state can therefore be written as **CurrentLocation = location**. The six possible states are L1, L2, L3, L4, L5, and L6.

E. Actions

An action is a movement from the current location to a directly connected location. Movement is allowed in both directions along every listed connection.

From	To	Movement cost
L1 Main Gate	L2 Administration Block	1 km
L2 Administration Block	L3 Library	2 km
L2 Administration Block	L4 Science Laboratory	2 km
L1 Main Gate	L6 Cafeteria	3 km
L4 Science Laboratory	L5 Student Affairs	2 km
L6 Cafeteria	L5 Student Affairs	2 km

F. Goal Test

The search has reached the goal when the student's current location is the required destination. For this assignment, the goal test is:

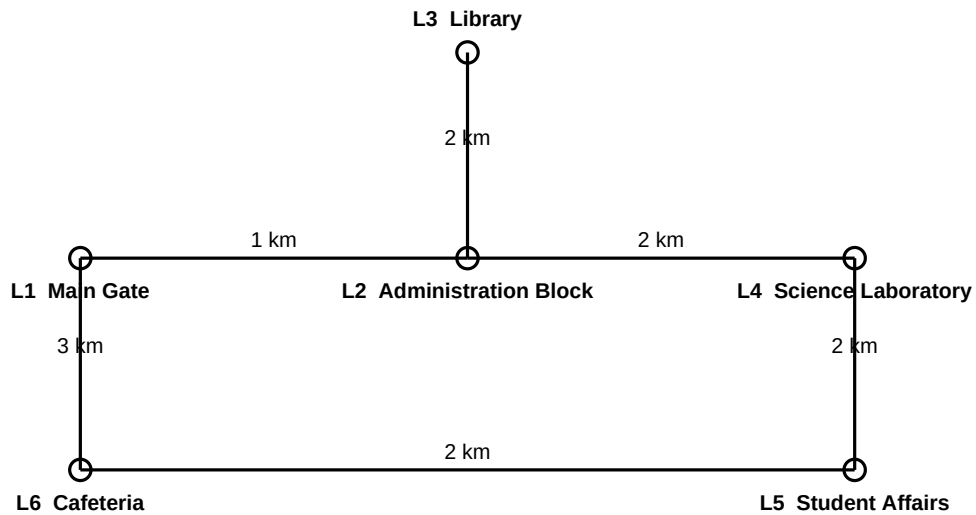
CurrentLocation == GoalLocation

G. Cost

The cost of an action is the distance of the connection travelled. The total path cost is the sum of the individual connection costs along the route. Because movement is bidirectional, each listed connection has the same cost in either direction.

H. State-Space Graph

The state-space graph below represents the six campus locations and the allowed movements. Nodes represent states (locations), while edges represent possible actions. The edge labels show movement costs.



Commit checkpoint: Part 1: problem definition